CLOUD SECURITY — PRACTICAL-BASED LEARNING

PBL 1: Deploy a Public Web Server on a Linux EC2 Instance

Standalone practical report

1. Aim

Deploy a Linux-based Amazon EC2 web server that displays the student's last four PRN digits and is accessible over HTTP from an external network.

2. Problem Statement

Launch a Linux EC2 instance, configure secure administrative access, install and start a web server, create a webpage displaying the required PRN suffix, and verify public HTTP access.

3. Tools and Technologies

- AWS Management Console and Amazon EC2
- Amazon Linux 2023
- Apache HTTP Server (httpd)
- SSH and an EC2 key pair (.pem)
- EC2 Security Groups
- HTML and curl

4. Procedure

Step 1 — Launch the EC2 instance

Launch a Linux EC2 instance, select or create a key pair, and record its current public IPv4 address. Confirm the instance is running before attempting SSH.

Step 2 — Configure the security group

- Allow SSH (TCP 22) only from the authorized client’s public IP using /32 notation.
- Allow HTTP (TCP 80) from the internet only if the assignment requires a publicly accessible website.

Do not expose SSH to all IPv4 addresses. The client’s local private IP is generally not the correct source for an internet-facing SSH rule.

Step 3 — Connect using SSH

```
ssh -i "key-name.pem" ec2-user@PUBLIC_IP
```

Replace the placeholder with the actual key filename and current public IP. Use the correct username for the selected Amazon Linux image.

Step 4 — Install and start Apache

```
sudo dnf install -y httpd\nsudo systemctl enable --now httpd\nsudo systemctl status httpd
```

Confirm that the service is active before creating the webpage.

Step 5 — Create the webpage

```
echo '<h1>PRN: LAST_FOUR_DIGITS</h1>' | sudo tee /var/www/html/index.html
```

Replace LAST_FOUR_DIGITS with the correct last four digits of the student's PRN.

Step 6 — Test the website

```
curl http://localhost
```

Then open http://PUBLIC_IP in an external browser. A local curl test alone does not prove public accessibility.

5. Troubleshooting History and Solutions

SSH connection timed out

An SSH timeout was encountered. Check that the instance is running, use its current public IP, verify the SSH inbound rule uses the client's current public IP/32, and confirm the instance has a valid public route and reachable network interface.

Windows private-key permissions

Windows SSH may reject a .pem file if its permissions are too broad. Restrict access to the current user and administrators, then retry SSH. Example PowerShell commands:

```
icacls "key-name.pem" /inheritance:r\nicacls "key-name.pem" /grant:r "$($env:USERNAME):(R)"
```

Use the actual path to the key file. Do not share or upload the private key.

Website not accessible

If the browser cannot load the page, verify Apache is active, confirm the HTML file exists, allow inbound TCP 80 in the instance security group, and use the current public IP. These are troubleshooting checks; the available history does not establish that every check was completed successfully.

6. Result

The intended configuration is a public Linux EC2 web server serving a page with the last four PRN digits. The available history records SSH timeout and key-permission troubleshooting, but does not establish a verified successful external browser test showing the correct PRN suffix.

FINAL STATUS: PARTIALLY VERIFIED. Mark the website as successfully deployed only after capturing the external browser page and confirming the displayed PRN suffix.

7. Security Measures

- Restrict SSH to the authorized public IP/32.
- Keep the private key confidential and apply restrictive file permissions.
- Open only the ports needed for the practical.
- Use a simple static webpage and avoid displaying sensitive personal information beyond the required identifier.
- Monitor AWS costs and stop or delete resources after the lab.

8. Conclusion

This practical covers launching a Linux EC2 instance, configuring SSH and HTTP security-group rules, installing Apache, and publishing a static webpage. SSH connectivity and key permissions were troubleshooting points. External website availability and the correct PRN suffix should be confirmed with a browser screenshot before claiming full completion.

9. Screenshots Checklist

- EC2 instance state and current public IPv4 address.
- Security group showing SSH restricted to authorized public IP/32 and HTTP on port 80.
- SSH terminal session.
- Apache service status.
- HTML file contents or successful local curl response.
- External browser displaying the webpage and correct PRN suffix.
- Cleanup or stopped/terminated resources, where applicable.